

CS 599 Final Report: Panoramic Generation from Perspective Video with Camera Motion

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Abstract

Panoramic video provides a distinct form of immersive control: viewers can choose where to look at each moment while following the captured camera trajectory. This makes camera motion central to the representation, since a moving panorama allows viewers to look around from multiple positions along a path rather than from a single anchor viewpoint. Generating such videos from regular perspective input is challenging however, as most viewing directions are never observed, and existing panoramic video models are often limited by anchor-view assumptions or noisy panoramic data when the camera is moving. To avoid training directly on such data, we study panoramic outpainting from perspective videos with camera motion using a training-free, geometry-grounded framework. Our method reconstructs observed content as 3D Gaussian Splats (3DGS), completes missing equirectangular regions with a panoramic image prior, lifts the completed panorama back into 3D, and renders the updated scene along the input trajectory. Experiments compare visual quality and input-view consistency against panoramic generation baselines, suggesting that image-based completion with 3D propagation is a promising direction for moving-camera panoramic video generation.

1. Introduction

Panoramic video is not simply video with a wider field of view. It changes the viewing experience by fixing the camera trajectory while leaving the viewer’s orientation open. Instead of inheriting the single gaze direction chosen by a perspective camera, the viewer can look around at each moment and decide which part of the surrounding scene to attend to. This makes panoramic video a useful intermediate representation for immersive media: it keeps the fidelity and temporal structure of RGB video while adding rotational control over the captured world.

This control is most valuable when the camera moves.

A static panoramic image only allows exploration from one point; a moving panoramic video allows exploration from many points along a path through the scene. The viewer is still constrained by the recorded trajectory, but is no longer constrained by the narrow perspective frustum selected at capture time. To realize this potential, panoramic generation methods must handle camera motion while preserving the geometry and temporal continuity of the scene.

Current methods only partially realize this goal. Some focus on anchor-view or weakly moving settings, while recent video-to-panorama systems train directly on large collections of 360-degree video [4, 12, 13, 16, 24, 25]. However, clean moving-camera panoramic video is difficult to collect: unlike tripod footage, it must be captured by a person, car, drone, or robot, which can introduce carrier artifacts, unstable motion, viewpoint transitions, and low-resolution training examples. Direct panoramic video training is also very expensive, as every frame covers the full viewing sphere and therefore must remain high resolution to feel immersive.

We study panoramic outpainting from perspective videos with camera motion. Rather than training a direct panoramic video model, we separate high-quality panoramic completion from geometry-based propagation. Our pipeline reconstructs observed content as a 3DGS scene [9, 11], renders a masked equirectangular reference panorama, completes missing directions with a panoramic image outpainting model [5], lifts the generated regions back into 3D, and renders the result along the input trajectory.

2. Related Work

2.1. Panoramic Image Outpainting

Panoramic image generation and outpainting aim to synthesize a complete equirectangular view from partial visual context or text conditioning. Compared with panoramic video generation, the image setting benefits from stronger generative priors, higher effective resolution, and cleaner training data. This mirrors the broader progress of vi-

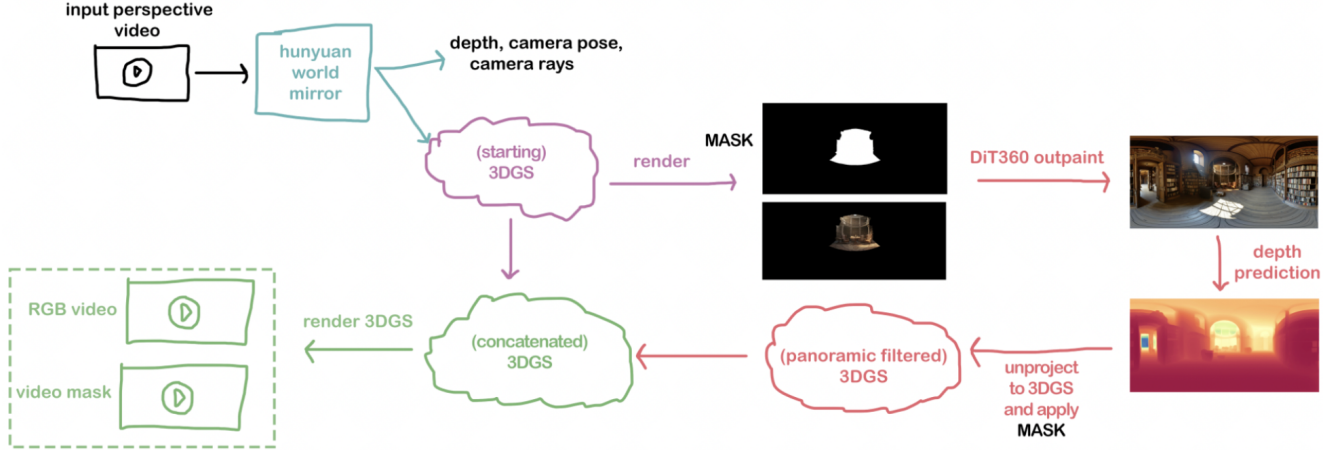


Figure 1. Overview of the proposed panoramic outpainting pipeline. A feed-forward 3D reconstruction stores observed content, a panoramic image model completes unobserved directions, and the completed content is lifted back into a renderable 3DGS representation before rendering the final panoramic video.

sual generation, where image models often become visually reliable before their video counterparts. For our setting, this distinction is important: the missing content around a perspective video frame should be high quality, seamless across the equirectangular boundary, and faithful to the observed region. Prior panoramic image methods study text-driven panorama synthesis, 360-degree outpainting, and seam-aware diffusion mechanisms [3, 5, 6, 23]. We therefore use a panoramic image outpainting model as a completion prior, while relying on geometry to propagate the completed content through time.

2.2. Panoramic Video Generation

Panoramic video generation extends image-level generation to temporally coherent sequences. However, many methods are most effective in fixed or anchor-view settings, where the camera center is assumed to remain approximately stable [4, 13, 16, 25]. Moving-camera panoramic video is harder because the model must learn how 3D scene structure projects into equirectangular frames over time. Large-scale video-to-panorama models such as Argus [12] and 360Anything [24] address this by mining substantial amounts of panoramic video data, but such data is often noisy: it may be low resolution, contain transitions or stabilization artifacts, and include visible carriers such as people, cars, or drones. Our work targets the same moving-camera goal, but avoids training a direct panoramic video generator by using a geometry-grounded image completion pipeline.

2.3. 3D Geometry and Scene Reconstruction

3D geometry provides a natural mechanism for view-consistent generation. Classical and learned reconstruction methods estimate camera motion, depth, or 3D structure from images and videos, making it possible to store

observations in a scene-centered representation. Recent feed-forward geometry models such as DUST3R [19], VGGT [15], CUT3R [17], and π^3 [20] show that useful 3D structure can be predicted directly from sparse or streaming visual observations. At the same time, 3DGS has become a practical explicit representation for rendering view-dependent RGB content from a point-like scene representation [9]. Feed-forward 3DGS methods combine these ideas by predicting renderable splats without requiring slow per-scene optimization [1, 2, 8, 11]. Our method uses this style of feed-forward reconstruction as an initialization: the predicted geometry stores what the perspective video observed, and the generated panoramic content is added back into the same renderable representation.

3. Method

Given a perspective video, our goal is to synthesize an equirectangular panoramic video along the same camera trajectory. The pipeline can be split into three stages: (1) estimate sparse camera poses and reconstruct the observed scene as 3DGS. (2) choose a reference pose, render the reconstruction into an equirectangular image, and use the rendered support to define which pixels should be preserved versus outpainted. (3) estimate depth for the completed panorama, lift only the newly generated regions back into 3DGS, and render the updated scene along an interpolated camera path. Figure 1 summarizes the full process.

3.1. Reconstructing Observed Content

We begin by processing the input perspective video with a feed-forward reconstruction model, HunyuanWorld-Mirror [11]. The input video is sampled into a sparse set of frames, and the model predicts camera poses, intrinsics,

dense 3D structure, and an initial 3DGS representation of the observed scene. We interpolate the estimated camera trajectory after reconstruction, ensuring the final panoramic video follows the input motion while also being rendered at any desired output FPS.

This reconstruction serves as a persistent memory of the observed scene. Instead of treating each perspective frame independently, visible content is stored in the shared 3D representation, allowing it to be queried from different viewing directions and updated with newly generated content. To determine what should be preserved versus generated, we render this 3DGS scene from a reference pose into an equirectangular panorama, along with alpha and depth support. When no reference frame is specified, we automatically select a reference pose with low rendered panoramic coverage, asking the outpainting model to fill directions that are poorly covered by the input trajectory. The rendered panorama provides the visible content already explained by the input video, while the support maps define which panoramic regions are observed and which remain unobserved. These unobserved regions become the target for panoramic outpainting in the next stage.

3.2. Completing Unobserved Panoramic Regions

The initial 3DGS scene represents only the parts of the scene observed by the input perspective video. To complete the surrounding view, we turn this partial reconstruction into a panoramic outpainting input and generate new observations for the missing directions.

Given the selected reference pose, we render the current 3DGS scene into an equirectangular image and compute a keep mask indicating which pixels are supported by observed content. We also render depth and alpha support from the reconstruction; the depth map is later used to align the generated panorama to the reconstructed scene scale, while the alpha support defines which regions should be preserved during outpainting. To avoid conditioning on unreliable geometry, we filter weakly supported or overly diffuse regions from the keep mask before passing it to the outpainting model.

We then adapt DiT360 [5], a text-to-image panoramic generation model, into an image-conditioned outpainting model. Given the rendered equirectangular image and keep mask, we first invert the input region into the diffusion latent space. During sampling, we use a scene prompt to guide generation and a keep mask to preserve the input source region while synthesizing unobserved regions. We tune the insertion time τ to control input preservation, and apply key scaling so larger outpainted regions better attend to smaller observed regions. The result is a completed panorama that preserves input regions from the reconstructed 3DGS scene while synthesizing missing regions with the visual quality and spherical consistency of DiT360.

3.3. Updating and Rendering the Panoramic Scene

After outpainting, the completed panorama is still only a 2D image. To make it useful for a moving panoramic video, we lift the newly synthesized regions back into the 3D scene. We estimate panoramic depth using a depth model such as MoGe [18] or DAP [10], optionally with a validity mask for sky or unreliable regions. Since monocular depth may not match the reconstruction scale, we align it to the rendered point-cloud depth in the overlap between generated depth, valid depth support, and the observed keep mask.

Only pixels outside the original rendered mask are lifted back into the scene. For each new RGB-D pixel, we cast an equirectangular ray from the reference camera pose, place a splat at the corresponding 3D point, and set its local scale from the angular footprint of an equirectangular pixel at that depth. The resulting generated splats are merged with the original 3DGS reconstruction, so observed content remains anchored by the input video while unobserved directions are supplied by the panoramic image prior.

Finally, the updated 3DGS scene is rendered as an equirectangular video along the interpolated camera trajectory. The renderer also outputs alpha and depth maps, which identify low opacity regions that remain after 3D lifting. For the filled variant of our method, we apply classical Telea inpainting [14], and the pipeline also supports optional VACE refinement with Wan [22] for video-based hole filling. The final output is an equirectangular panoramic video that preserves the input scene while following the input camera trajectory.

4. Experiments

4.1. Experimental Setup

We evaluate panoramic generation from RGB perspective videos with camera motion. Our test inputs are perspective clips from a subset of the Argus test set, where the goal is to generate an outpainted equirectangular video from the input perspective clip. We compare against panoramic generation baselines including Viewpoint [4] and Argus [12], and report two variants of our method: *Ours Render*, the direct render from the updated 3DGS scene, and *Ours Filled*, which applies hole filling to low opacity regions in the rendered result.

4.2. Metrics

We follow the evaluation protocol used by Argus [12]. First, we measure consistency with the observed camera views by reprojecting generated panoramic videos back into the original perspective camera trajectory. This produces perspective clips that can be compared to the input or reference views using PSNR, SSIM [21], and LPIPS [26]. Second, we evaluate video-level perceptual quality using VBench [7]. Following Argus, we compute the VBench metrics on per-



Figure 2. Qualitative comparison for one representative frame from a moving-camera clip. We show perspective crops extracted from each panoramic frame in four viewing directions (left, front, right, back). The input perspective crop is represented by the blue box in the Ground Truth image

Method	Reprojection Consistency			VBench Quality		
	PSNR $\uparrow$	LPIPS $\downarrow$	SSIM $\uparrow$	Image $\uparrow$	Aesthetic $\uparrow$	Motion $\uparrow$
Viewpoint [4]	14.6967	0.6000	0.4158	0.3788	0.4178	0.9830
Argus [12]	15.8395	0.4999	0.4302	0.4006	<u>0.5222</u>	<u>0.9837</u>
Ours Render	<u>17.2710</u>	0.4397	<u>0.4671</u>	0.5098	0.5209	0.9823
Ours Filled	17.7467	<u>0.4421</u>	0.4749	<u>0.4902</u>	0.5323	0.9839

Table 1. Quantitative comparison on the Argus-derived evaluation set. Reprojection metrics measure consistency with the original camera views, while VBench metrics summarize perceptual video quality on cube-face crops.

spective cube-face crops from the generated panorama, using only the left, front, right, and back directions when calculating metrics.

4.3. Quantitative Results

Table 1 reports quantitative results on the Argus-derived evaluation set. Our method improves both reprojection consistency and VBench quality over both baselines. *Ours Filled* achieves the best PSNR, SSIM, Aesthetic Quality, and Motion Smoothness, while the direct render achieves the best LPIPS and Imaging Quality. Overall, our method performs better quantitatively than the baselines.

4.4. Qualitative Results

We also show qualitative results to illustrate visual artifacts and scene consistency differences that are not fully captured by aggregate metrics. In particular, these examples highlight whether methods preserve observed content, generate plausible unseen regions, and avoid localized panoramic artifacts such as unnatural geometric distortion or hallucinated carrier-objects like humans. Figure 2 shows a representative examples where our output is compared against baselines and the ground truth.

5. Conclusion

In this paper, we introduced a training-free framework for panoramic generation from perspective videos with camera motion. Our approach is motivated by the observation that panoramic image priors are well suited for synthesizing missing directions, while a 3DGS scene representation provides a natural way to store observed and generated content for rendering along the input camera trajectory. By anchoring generation to observed 3D scene content, our method improves input-view consistency, benefits from the visual quality of panoramic image models, and avoids common pitfalls in direct data-driven panoramic video generation.

The main limitations of our current framework also point to natural directions for future work. Remaining holes and artifacts can appear in difficult unobserved regions, suggesting the need for stronger refinement or iterative panoramic completion. The static 3DGS representation also limits performance on dynamic scenes, where moving objects are not naturally modeled and may become blurred or inconsistently rendered over time. Finally, the current inference pipeline is relatively slow, motivating for more efficient implementations that produce longer and higher-resolution panoramic videos.

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